

# Yiheng Zhang

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[Website](#) | [Github](#) | [Linkedin](#)

## RESEARCH GOAL

Synthesize robust geometric processing method with generalizable learning method to empower traditional computer graphics domain. (See more details in my Research Proposal!)

Interests:

**3D Computer Vision:** Object & Scene-level Generation,

**Computer Graphics:** Geometry Processing and Geometric Learning: Operators & Quad,

**Agent:** Agentic Game/Movie Understanding & Design,

**Robotics:** Physics-based & Cognitive/Psychological Simulation

## EDUCATION

National University of Singapore(NUS)

M.S., Software Engineering

Aug. 2024 – Nov. 2025

Singapore

Nanjing University of Posts and Telecommunications

Bachelor of Science, Software Engineering

Bell Honors School (Elite Program), GPA: 3.88, Rank: 1st, Merit Student

Sept. 2020 – June. 2024

Nanjing, China

## EXPERIENCE

Research Scientist Intern

Tencent Games AI Center

Leader: RSR. Le Wan, Prof. Cheng Lin(MUST)

Sept. 2025 – Now

Shenzhen, China

Remote Research Intern

Intelligent Graphics Lab in Hong Kong University of Science and Technology

Advisor: Prof. Yuan Liu(HKUST), Prof. Wenping Wang(TAMU, ACM & IEEE Fellow)

Jun. 2025 – Now

Remote

Research Intern

Metaverse AI Interdisciplinary Innovation Lab in Tsinghua University

Advisor: RSR. Yuwang Wang, Prof. Qionghai Dai(Academician)

Oct. 2024 – Aug. 2025

Beijing, China

Summer Research Intern

Center for Graphics and Geometric Computing in Technion – Israel Institute of Technology

Advisor: Prof. Mirela Ben-Chen

Jul. 2024 – Aug. 2024

Haifa, Israel

## PUBLICATION

(Equal contribution are denoted by “\*”, click on the [blue line](#) to see the **detailed Project Page & code online.**)

### Conference

[C4] **Yiheng Zhang**, Tingrui Shen, Zhe Zhu, Zhuojiang Cai, Tianxiao Li, Rui Xu, Qiujie Dong, Jiepeng Wang, Zhiyang Dou, Le Wan, Yuwang Wang, Yuan Liu, Cheng Lin, Wengping Wang, “Hybridgen: autoregressive generation model for quad dominant mesh”, submitted to *SIGGRAPH*, 2026.

[C3] **Yiheng Zhang\***, Zhuojiang Cai\*, Mingdao Wang\*, Meitong Guo, Tianxiao Li, Li Lin, Yuwang Wang, “[M3DLayout: A Multi-Source Dataset of 3D Indoor Layouts and Structured Descriptions for 3D Generation](#)”, under review by *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

[C2] Tingrui Shen\*, **Yiheng Zhang\***, Chen Tang, Chuan Ping, Zixing Zhao, Le Wan, Yuwang Wang, Ronggang Wang, Shengfeng He, “[FlashMesh: Faster and Better Autoregressive Mesh Synthesis via Structured Speculation](#)”, under review by *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

[C1] Zhe Zhu, Le Wan, Rui Xu, **Yiheng Zhang**, Honghua Chen, Zhiyang Dou, Cheng Lin, Yuan Liu, Mingqiang Wei, “[PartSAM: A Scalable Promptable Part Segmentation Model Trained on Native 3D Data](#)”, under review by *International Conference on Learning Representations (ICLR)*, 2026.

## Journal

[J2] Zhuojiang Cai\*, **Yiheng Zhang\***, Meitong Guo, Mingdao Wang, Yuwang Wang, “[LSS3D: Learnable Spatial Shifting for Consistent and High-Quality 3D Generation from Single-Image](#)”, accepted by *MMSP*, 2025 (IF: 7.3).

[J1] Zhixun Liu\*, **Yiheng Zhang\***, Xinyao Han, Wanting Zhou, “[Animation Image Transfer Using CycleGAN](#)”, accepted by *ICCAID*, 2022.

## PROJECTS

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### **Tri2Quad: Geometry-Aware Triangle-to-Quad Mesh Conversion Operator**

Sept. 2025 – Oct. 2025

*Advisor: Prof. Cheng Lin, Prof. Yuan Liu, Prof. Wengping Wang*

- Developed a high-performance triangle-to-quad conversion operator, formulated as a maximum-weight matching problem on the triangle adjacency graph and solved with modified Blossom matchers for quad-dominant remeshing.
- Integrated rich geometric filters into candidate edge generation before optimization to improve solver efficiency and avoid poor merges, while additionally leveraging a unified graph-connectivity-based propagation scheme to guarantee globally post-merge consistent normals.
- The operator has been **the key data-curation step** of my HybridGen SIGGRAPH paper and **deployed in Tencent’s production pipeline**, achieving state-of-the-art quad quality compared with PyMeshLab, Blender’s built-in remesher, and Hunyuan’s ILP-based operator.

### **Hybrid Intrinsic–Extrinsic Spectral Operators for DiffusionNet**

July. 2024 – Aug. 2024

*Advisor: Prof. Mirela Ben-Chen*

- Replaced and complemented the standard Laplace–Beltrami operator in DiffusionNet with Steklov and elastic-basis operators, enabling the network to jointly encode intrinsic surface geometry and extrinsic embedding information.
- Demonstrated consistent improvements over the original DiffusionNet baseline on mesh segmentation, correspondence, and feature extraction benchmarks, showcasing a concrete case where classical geometry-processing operators and deep learning architectures are tightly integrated for better geometric understanding.

## ENTREPRENEURSHIP & LEADERSHIP

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### **Former CEO & Founder | Nanjing Ta’an Zhilian Technology Co., Ltd.**

Apr. 2023 – Mar. 2024

- \* Utilized some machine learning algorithms for predicting and hardware sensors for gathering data to develop a core tower crane warning system based on Unity3D, achieving synchronization between real and virtual tower cranes and significantly improving the accuracy of tower crane warnings and ensuring the safety of construction workers.
- \* Achieved contract with the construction projects of Jiangsu Academy of Agricultural Sciences with the contract amount of 5.75 million RMB, achieved the company’s account statement of 2 million RMB.
- \* Patents: First Inventor, A Tower Crane Safety Detection Method based on the Faster R-CNN Network.
- \* Software Copyrights: First Author, Tower Crane Safety Warning Software based on FlowFormer Model (2023SR0697679)
- \* First Author, Virtual Tower Crane Environment Generation Dataset System based on Reinforcement Learning, (2023SR0723714)

### **Team Leader | Global Business Leadership Program, Stanford University**

Jan. 2024 – Feb. 2024

- \* Received a full scholarship to participate in Global business Leadership program at Stanford University and UC Berkeley.
- \* Participated in Global Innovation Plan Competition at Intel’s headquarters in Silicon Valley as the team leader, winning first place and securing an investment prize and continued interest from investors.

### **Class representative | SE Class, NUS**

Sep. 2024 – Nov. 2025

### **Director | Secretariat, BELL Honors College, NJUPT**

Oct. 2022 – Oct. 2023